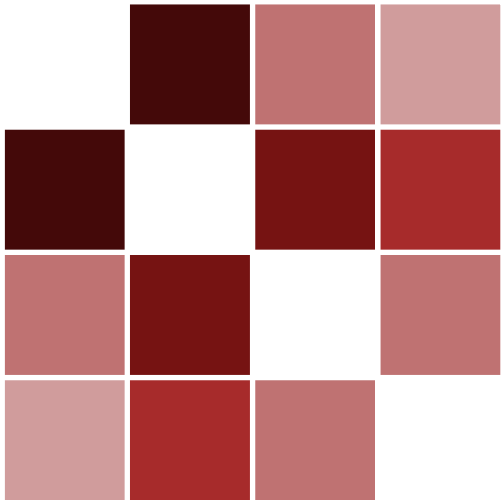






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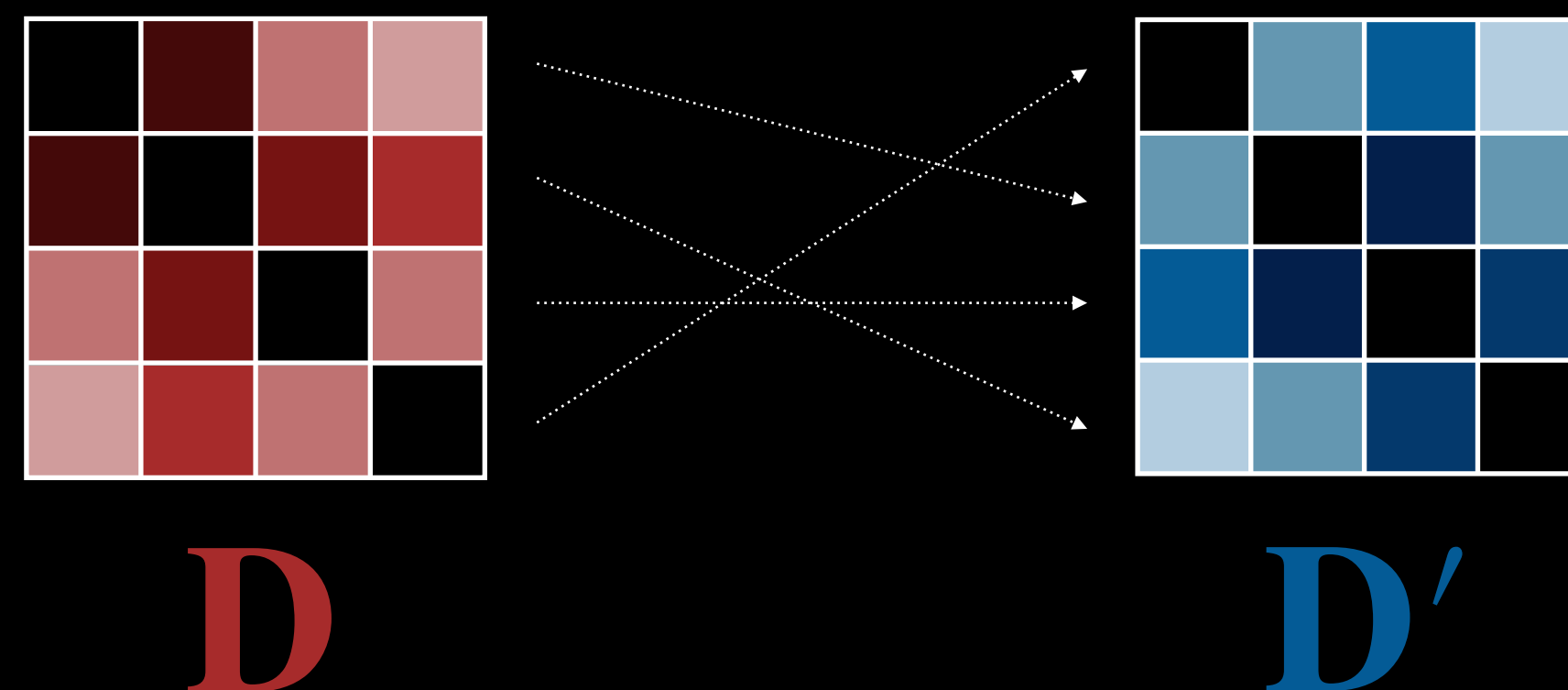
Gromov-Wasserstein: Discrete Case

- For discrete $\alpha = \sum_i \mathbf{a}_i \delta_{\mathbf{x}_i}$, $\beta = \sum_j \mathbf{b}_j \delta_{\mathbf{y}_j}$, let $\mathbf{D}_{ii'} = d_X(\mathbf{x}_i, \mathbf{x}_{i'})$ and $\mathbf{D}_{jj'} = d_Y(\mathbf{y}_j, \mathbf{y}_{j'})$.

Then:

$$\text{GW}^2((\mathbf{a}, \mathbf{D}), (\mathbf{b}, \mathbf{D}')) = \min_{\pi \in \Pi(\mathbf{a}, \mathbf{b})} \sum_{i, i', j, j'} |\mathbf{D}_{ii'} - \mathbf{D}'_{jj'}|^2 \pi_{ij} \pi_{i'j'}$$

- ~ finding an alignment that minimizes distances between distance matrices



Gromov-Wasserstein: Properties